Gennoor Tech
TRAIN. INNOVATE. build.

COURSE OUTLINE

Agentic Attack Surface Management

Discover, reconcile, contextualize and assign exposed assets

- 3 days (8 hours per day)
- Intermediate
- Instructor-led, hands-on labs
- Capstone project

About this course

Who should attend

External attack surface teams, security operations, cloud security, IT asset management, MSSPs.

Prerequisites

- DNS, HTTP and TLS fundamentals
- Asset inventory concepts
- Python
- Basic cloud and network knowledge

What you will be able to do

- Build a normalized asset inventory from approved sources with explicit scope enforcement
- Reconcile domains, certificates, cloud resources, services and ownership records
- Detect inventory changes and potential stale or orphaned assets
- Correlate assets with vulnerability and exposure findings, separating observed facts from inferred exposure
- Generate evidence-backed ownership and remediation tasks while blocking out-of-scope discovery

Tools and technologies

- Claude
- Python
- FastAPI
- PostgreSQL
- Certificate transparency
- DNS (fixtures)
- Cloud inventory (fixtures)
- Azure inventory (fixtures)
- Jira (mock)
- pytest
- Docker Compose

Table of contents

DAY 1 3 modules

MODULE 1

Authorization, scope policy and the lab

- Explicit allowlist of lab-owned domains and synthetic assets, exclusions, rate limits, audit trail
- Arbitrary target entry is disabled in the course environment
- Why weak scope is the biggest risk in attack surface tooling

HANDS-ON LAB

Implement the scope policy and prove that out-of-scope targets are blocked and logged.

DELIVERABLE

Scope policy with audit.

MODULE 2

Source adapters

- Fixture-based adapters for DNS records, certificate metadata, cloud inventory, CMDB entries and scanner findings
- Adapter contract, timeouts, retries and least privilege

HANDS-ON LAB

Build and test the five adapters against fixtures.

DELIVERABLE

Source adapter layer.

MODULE 3

Asset normalization and canonical IDs

- Canonical asset IDs; source, first and last seen, confidence, environment and relationships
- Provenance on every record

HANDS-ON LAB

Normalize all sources into the inventory store.

DELIVERABLE

Normalized asset inventory.

DAY 2 3 modules

MODULE 4

Entity resolution

- Deterministic signals: exact hostname, certificate SAN, cloud resource ID, verified ownership
- Uncertain matches go to a review queue, never silently merged
- Precision-oriented tests

HANDS-ON LAB
Implement entity resolution with a review queue and measure link precision.

DELIVERABLE
Entity resolution with review queue.

MODULE 5

Change detection

- Snapshot comparison: new, changed, missing, stale and potentially orphaned assets
- Synthetic fixtures with known expected outcomes

HANDS-ON LAB
Build snapshot comparison and validate it against the fixture set.

DELIVERABLE
Change detection engine.

MODULE 6

Exposure correlation with an agent

- Joining assets to scanner findings, service metadata and business criticality
- The agent summarizes exposure but must distinguish observed facts from inferred exposure
- Unsupported exposure claims are measured

HANDS-ON LAB
Correlate exposure and review the agent's output for unsupported claims.

DELIVERABLE
Exposure correlation.

DAY 3 3 modules

MODULE 7

Ownership resolution

- Synthetic CMDB records, tags and repository metadata to suggest a responsible team
- Evidence attached to every suggestion; manual correction supported

HANDS-ON LAB
Resolve ownership for the inventory and correct three wrong suggestions.

DELIVERABLE
Ownership resolver.

MODULE 8

Prioritization, ticketing and inventory quality metrics

- Transparent ranking criteria and review tickets through a mock connector
- Entity-link precision, unowned asset rate, stale-asset classification, change-detection accuracy
- Asset graph, reports and audit

HANDS-ON LAB
Rank, raise review tickets and publish the quality metrics.

DELIVERABLE
Metrics report and asset graph.

MODULE 9

Capstone: scoped attack surface platform

- Approved sources to owned, prioritized exposures in one run
- Acceptance criteria, limitations, demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end attack surface management lab.

Capstone

Capstone project

A platform that consolidates approved asset sources, detects changes and inventory gaps, links assets to owners and findings, and prioritizes exposure review, with every discovery action constrained to authorized scope.

Assessment

Assessment area	Weight	Evidence
Working functionality	**25%**	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	**20%**	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	**20%**	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	**15%**	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	**10%**	Traceable findings, sources, assumptions, logs
Documentation and presentation	**10%**	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.